

Introduction to Predictive Modeling

From Classical Regression and Classification to Neural Networks from Scratch

by Dr. Rishikesh Yadav

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Assistant Professor, School of Mathematical and Statistical Sciences, IIT Mandi, India

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Course Objectives

The objectives of this lecture would be:

- Learn the basics of **regression** and **classification** techniques.
- Train **regression** and **classification** using **the gradient descent** algorithm.
- Understand key ideas from scratch in **neural networks** and **deep learning**.
- Build and train simple **neural networks** from scratch and also using **TensorFlow**.
- Bridge **traditional statistical regression methods** with **machine learning algorithms**.

Note: Throughout the whole workshop lectures, we will be only covering the **supervised learning** task (that means we will have input/output **(labeled)** data). We will not cover any unsupervised, semi-supervised, or reinforcement algorithms.

Some Good References

- Bishop, C. M. (2007). *Pattern Recognition and Machine Learning*. Springer.
- Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (3rd ed.). O'Reilly. [[GitHub companion code](#)]
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press. [[Free online](#)]
- Chollet, F. (2021). *Deep Learning with Python* (2nd ed.). Manning.
- Nielsen, M. A. (2015). *Neural Networks and Deep Learning*. [[Free online book](#)]
- Online learning platforms:
 - [Coursera - Deep Learning Specialization](#)
 - [fast.ai - Practical Deep Learning for Coders](#)
 - [DeepLearning.AI](#)

Predictive Modeling

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Why Predictive Modeling (1)?

While exploring the data, why are just summary statistics and graphical summaries not enough?

Think About:

- What about prediction?
- What about uncertainty quantification?
- What can we say when we have another set of data from the same population?

All of them can be solved by performing statistical/machine learning modeling.

Why Predictive Modeling (2)?

- In data science and applied statistics, we often aim to understand, quantify, and **predict** outcomes based on observed variables.
- Two fundamental types of predictive tasks:
 - **Regression**: Predict a **continuous** outcome (e.g., income, temperature, recovery time)
 - **Classification**: Predict a **categorical** outcome (e.g., disease status, pass/fail)
- These models allow us to:
 - Forecast outcomes
 - Evaluate importance of features
 - Make data-driven decisions
- Commonly applied in healthcare, business, environmental science, social sciences, and more.

Common Terminology in Predictive Modeling

- **Target (label, output, response):**
 - The variable we aim to predict.
 - Also called: **dependent variable** (in statistics), **output variable**, **ground truth**.
 - Can be:
 - **Continuous** → regression (e.g., price, temperature)
 - **Categorical** → classification (e.g., spam vs. not, disease yes/no)
- **Features (inputs, attributes, predictors):**
 - The variables used to predict the target.
 - Also called: **independent variables**, **explanatory variables**, **covariates** (in statistics).
 - Represent measurable characteristics or properties of the data.
- **Example - Predicting Diabetes Diagnosis (Classification):**
 - **Target:** Diabetic (Yes = 1, No = 0)
 - **Features:** Age, BMI, glucose level, blood pressure
- **Notation:**
 - **X:** Feature matrix and **Y:** Target vector

Regression vs. Classification

- **Regression:**

- Predicts a **continuous numeric** response variable.
- **Example:** Predicting a person's blood pressure based on age and weight.

- **Classification:**

- Predicts a **categorical or binary** class label.
- **Example:** Predicting whether a tumor is benign or malignant based on imaging features.

- **Shared foundation:**

- Both use observed variables (predictors) to model outcomes.
- Both aim to learn a function $f(\mathbf{X})$ to map inputs to outputs.

General Formulation of Predictive Modeling

- We aim to learn the relationship between predictors $\mathbf{X}$ and a response Y :

$$Y = f(\mathbf{X}) + \varepsilon$$

- For **regression**:
 - $Y \in \mathbb{R}$: continuous output
 - E.g., house price, blood pressure
- For **classification**:
 - $Y \in \{0, 1\}$ or more general classes
 - E.g., fraud detection, disease status
- $f(\mathbf{X})$ may be:
 - Linear (e.g., linear regression, logistic regression)
 - Non-linear (e.g., trees, neural networks)
- Error ε is the one that kind of make things and models accounting for uncertainty.

Example 1: Regression Task — Housing Prices

- **Goal:** Predict house prices using features of the property.
- **Target (Y):** Sale price (continuous)
- **Predictors (X):**
 - Square footage
 - Number of bedrooms
 - Location
 - House age
- **Key Questions:**
 - How does location affect price?
 - What is the price increase per added bedroom?

Example 2: Classification Task — Medical Study

- **Goal:** Predict patient outcome (e.g., disease presence).
- **Target (Y):** Treatment success (Yes/No)
- **Predictors (X):**
 - Age
 - Sex
 - Treatment type
 - Comorbidities
- **Key Questions:**
 - Which features are most predictive of outcome?
 - Can we flag high-risk patients before treatment?

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Linear Regression (1)

- **Linear Regression** is a statistical method for modeling the relationship between a dependent variable and one or more independent variables.
- The relationship is modeled using a linear predictor function whose unknown parameters are estimated from the data.
- In its simplest form, with one dependent variable y and one independent variable x , the linear regression model, so-called **Simple Linear Regression (SLR)** is:

$$y = w_0 + w_1x + \varepsilon = \mathbf{w}^T \mathbf{x} + \varepsilon, \quad \mathbf{x} = (1, x)^T, \quad \mathbf{w} = (w_0, w_1)^T,$$

where:

- y is the output variable.
- x is the input variable.
- w_0 is the intercept term, which represents the expected value of y when $x = 0$.
- w_1 is the slope term, which represents the change in y for a one-unit change in x .
- ε is the error term, which accounts for the variability in y that

Linear Regression (2)

- In a classical regression setting, the parameters w_0 and w_1 are estimated from the data using methods such as **Ordinary Least Squares (OLS)**, which minimizes the sum of the squared differences between the observed values and the predicted values.
- The **OLS** estimate of the parameters can be obtained by solving the following normal equations:

$$\begin{pmatrix} \hat{w}_0 \\ \hat{w}_1 \end{pmatrix} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

where:

- $\mathbf{X}$ is the design matrix, including a column of ones for the intercept and a column of the predictor variable values.
- $\mathbf{y}$ is the vector of observed values of the dependent variable.
- SLR can be extended to include multiple independent variables, resulting in **Multiple Linear Regression (MLR)**:

$$y = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_p x_p + \epsilon.$$

where $x_1, x_2, \dots, x_p$ are the predictor variables.

Interpretation of Regression Model Parameters

- **Estimated (Predicted) Regression Line:**

$$\hat{y} = \hat{w}_0 + \hat{w}_1 x$$

- **Interpretation of $\hat{w}_1$ (Slope):**

- Represents the estimated change in y for a one-unit increase in X .
- **Example:** If $\hat{w}_1 = 2$, then for each additional unit of X , Y is expected to increase by 2 units.

- **Interpretation of $\hat{w}_0$ (Intercept):**

- Represents the estimated value of Y when $X = 0$.
- **Note:** The intercept may not always have a meaningful interpretation, especially if $X = 0$ is outside the range of observed data.

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What is Classification?

- **Classification** is a supervised learning task where the goal is to predict a **categorical target variable** (label).
- The output is a class label rather than a continuous number.
- **Binary classification**: Two classes (e.g., spam vs. not spam, disease vs. no disease)
- **Multiclass classification**: More than two classes (e.g., handwritten digit recognition: 0 to 9)
- Given input features $\mathbf{x}$, the goal is to learn a function that predicts class membership:

$$f(\mathbf{x}) \in \{0, 1\} \quad (\text{or more generally, } \{1, 2, \dots, K\})$$

- **Example**: Predicting if a patient has diabetes based on age, BMI, and glucose levels.

Logistic Regression (1)

- **Logistic Regression** is a classical statistical model used for **binary classification**.
- The model estimates the **probability** that a given input belongs to the positive class (e.g., $Y = 1$).
- The relationship between the predictors $\mathbf{x}$ and the probability $P(Y = 1 \mid \mathbf{x})$ is modeled as:

$$P(Y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}}}$$

- Here, $\sigma(\cdot)$ is the logistic (sigmoid) function.
- Outputs are interpreted as class probabilities, and a threshold (e.g., 0.5) is used to make class predictions.

No Closed-Form Solution for Logistic Regression

- Unlike linear regression, logistic regression does **not** have a closed-form solution for estimating parameters $\mathbf{w}$.
- Instead, we use an **optimization approach** to maximize the likelihood of the observed data.
- This involves minimizing the **log-loss** (or cross-entropy loss), given by:

$$\mathcal{L}(\mathbf{w}) = - \sum_{i=1}^n [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$$

where $\hat{p}_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$.

- Since this is a convex optimization problem, we use **iterative algorithms** such as **gradient descent** to find the optimal $\mathbf{w}$.

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Gradient Descent Algorithm (1)

Gradient Descent Algorithm is an optimization technique used to minimize a loss function by iteratively moving towards the minimum of the function.

The main components of the gradient descent algorithm are:

- **Loss Function:**

- The loss function measures the error or difference between the predicted values and the actual values. That means it measures how well the model's predictions match the actual data.
- The goal of gradient descent is to minimize this loss function to improve the accuracy of the model.
- Examples of loss functions include **Mean Squared Error (MSE)** for regression problems and **Cross-Entropy Loss** for classification problems.
- Mathematically, for a set of predictions $\hat{y}$ and actual values y , the Mean Squared Error is given by:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

Gradient Descent Algorithm (2)

- **Gradient:**

- The gradient is a vector of partial derivatives of the loss function with respect to all the parameters of the model.
- It points in the direction of the steepest ascent of the function.
- In gradient descent, we move in the opposite direction of the gradient to find the minimum of the loss function.
- For a parameter $\mathbf{w} = (w_0, w_1, \dots, w_p)^\top$, the gradient of the loss function $\mathcal{L}$ with respect to $\mathbf{w}$ is given by:

$$\nabla_{\mathbf{w}}\mathcal{L} = \left(\frac{\partial \mathcal{L}}{\partial w_0}, \frac{\partial \mathcal{L}}{\partial w_1}, \dots, \frac{\partial \mathcal{L}}{\partial w_p} \right).$$

- **Update Rule:**

- In each iteration of gradient descent, the model parameters are updated using the gradient and the learning rate η .
- The update rule for a parameter $\mathbf{w}$ is given by:

$$\mathbf{w} := \mathbf{w} - \eta \nabla_{\mathbf{w}}\mathcal{L},$$

where η is the learning rate and $\nabla_{\mathbf{w}}\mathcal{L}$ is the gradient of the loss function with respect to $\mathbf{w}$.

Gradient Descent Algorithm (3)

- **Learning Rate η :**

- The learning rate is a hyperparameter that determines the size of the steps taken towards the minimum of the loss function.
- A smaller learning rate means smaller steps, leading to more accurate convergence but slower progress.
- A larger learning rate can speed up the process but risks overshooting the minimum and potentially diverging.
- It is important to choose an appropriate learning rate to ensure efficient and effective training of the model.

- **Convergence:**

- The algorithm iteratively updates the parameters until the loss function converges to a minimum value.
- Convergence can be determined by setting a threshold for the change in the loss function or by specifying a maximum number of iterations.

Gradient Descent Algorithm: Algorithm Steps

- **Initialize:** Start with an initial guess for the parameters.
- **Compute Gradient:** Calculate the gradient of the loss function with respect to each parameter.
- **Update Parameters:** Adjust the parameters by moving them in the direction opposite to the gradient, scaled by the learning rate.
- **Iterate:** Repeat the process until the parameters converge to the minimum or a stopping criterion is met.

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Lab 1: Gradient Descent for Logistic Regression

- **Recall:** **Gradient descent** is an iterative method to find parameters that minimize a loss function.
- **Loss function** (Binary Cross-Entropy / Log-Loss):

$$\mathcal{L}(\mathbf{w}) = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)],$$

where $\hat{y}_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$ and $\sigma(z) = \frac{1}{1+e^{-z}}$.

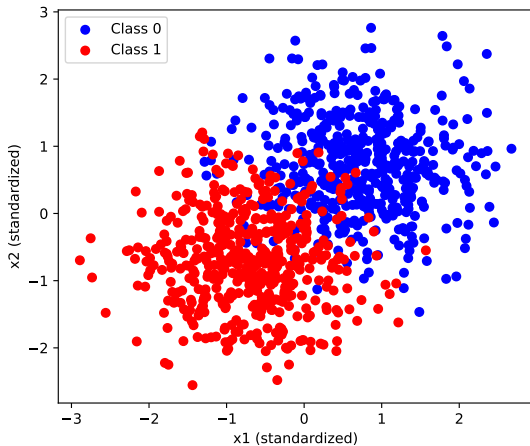
- **Gradient:**

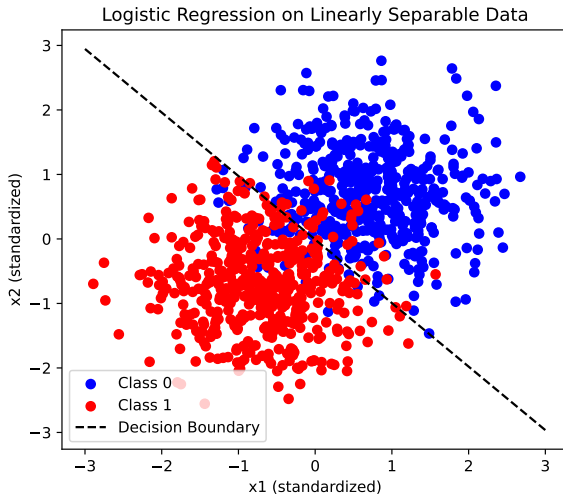
$$\nabla \mathcal{L}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \mathbf{x}_i$$

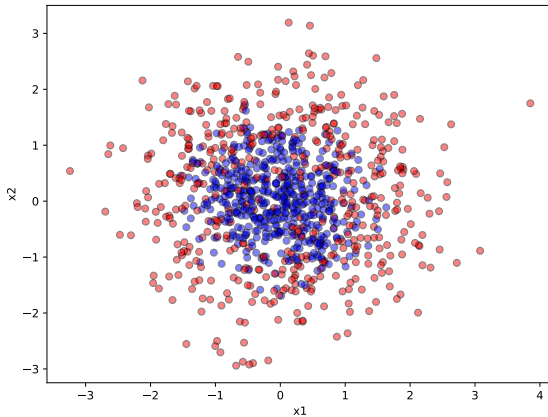
- **Parameter update rule:**

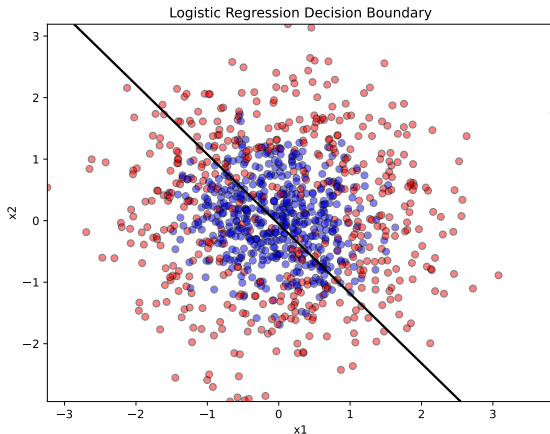
$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \cdot \nabla \mathcal{L}(\mathbf{w}^{(t)})$$

where η is the learning rate.









1

¹Open the python code from the GitHub Link:

https://github.com/yadavrishikesh/Masterclass-IITMandi/blob/main/Module3/code/logistic_regression_from_scratch.ipynb

Lab 2: Gradient Descent for Linear Regression

- **Loss function** (Mean Squared Error, MSE):

$$\mathcal{L}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad \hat{y}_i = w_0 + w_1 x_i$$

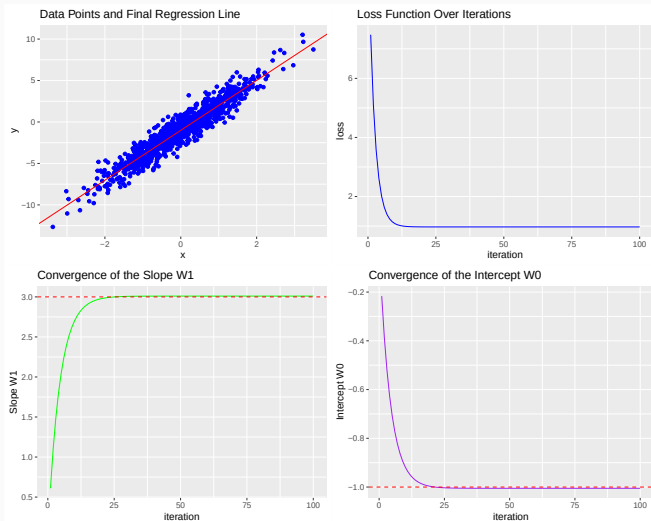
- **Gradients:**

$$\frac{\partial \mathcal{L}}{\partial w_1} = -\frac{2}{n} \sum_{i=1}^n x_i (y_i - \hat{y}_i), \quad \frac{\partial \mathcal{L}}{\partial w_0} = -\frac{2}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

- **Parameter updates:**

$$w_1 \leftarrow w_1 - \eta \frac{\partial \mathcal{L}}{\partial w_1}, \quad w_0 \leftarrow w_0 - \eta \frac{\partial \mathcal{L}}{\partial w_0}$$

Results Lab 2



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²Open the python code from the GitHub Link:

https://github.com/yadavrishikesh/Masterclass-IITMandi/blob/main/Module3/code/gradient_descent_LM.ipynb

Exercise: Implement the Gradient Descent for Polynomial Regression

- **Model:** $y = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M + \epsilon$
- **Predicted output:**

$$\hat{y}_i = \sum_{j=0}^M w_j x_i^j$$

- **Loss function** (Mean Squared Error):

$$\mathcal{L}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- **Gradient for each weight** w_j :

$$\frac{\partial \mathcal{L}}{\partial w_j} = -\frac{2}{n} \sum_{i=1}^n (y_i - \hat{y}_i) x_i^j$$

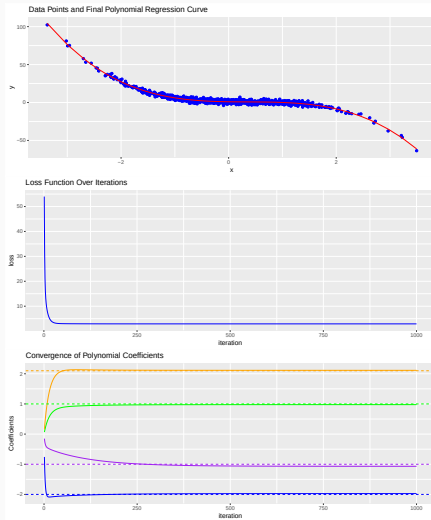
- **Update rule:**

$$w_j \leftarrow w_j - \eta \cdot \frac{\partial \mathcal{L}}{\partial w_j}, \quad \text{for } j = 0, \dots, M$$

Exercise: Implement the Gradient Descent for Polynomial Regression

Look at the exercise from https://github.com/yadavrishikesh/Masterclass-IITMandi/blob/main/Module3/assignments/gradient_descent_assignment.ipynb

Results Exercise



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³Open the python code from the GitHub Link:

https://github.com/yadavrishikesh/Masterclass-IITMandi/blob/main/Module3/code/gradient_descent_polynomial.ipynb

Regression vs. Classification Summary

Aspect	Regression	Classification
Output type	Continuous	Categorical
Model example	Linear / Polynomial Regression	Logistic Regression
Loss function	$\mathcal{L}_{\text{MSE}} = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$	$\mathcal{L}_{\text{log}} = -\frac{1}{n} \sum [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]$
Closed-form solution	Yes (e.g., OLS)	No
Estimation method	Matrix inversion or GD	Gradient descent
Goal	Predict a number	Predict a class or probability

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Model Validation: Why Do We Need It?

How do we know if our model is actually any good?

- Is a **linear regression model** good enough for a given data, OR
- Do we need a **polynomial model**? If so, what **degree** is enough?
- Are any models other than **polynomial** needed to be looked at?
- A model that performs well on **training data** may fail on **new data**.
- We need to assess the model's **generalization performance**.
- Model validation helps avoid **overfitting** and supports better **model selection**.

Training, Validation, and Test Splits

- **Training Set** ($\approx 60\text{--}70\%$ of data): Used to **fit** the model - the model “**learns**” from this data.
- **Validation Set** ($\approx 15\text{--}20\%$):
 - Used to **tune hyperparameters** and monitor performance.
 - Allows techniques like **early stopping** or **model selection**.
- **Test Set** ($\approx 15\text{--}20\%$):
 - Held out until **final evaluation**.
 - Measures true **generalization** to **unseen data**.

Important Note: In most cases, we only divide the whole data into training and testing datasets.

Most Common Data Splitting Strategies

- **Random Split:**
 - Common and simple.
 - Risky if the dataset is small or imbalanced.
- **Stratified Split:**
 - Preserves class proportions in classification.
- **Time-Based Split:**
 - Used for time-series data where order matters.

Cross-Validation (CV)

- Splits data into multiple folds and rotates the validation set.
- Common types:
 - **k-Fold CV**: Divides data into k equal parts
 - **Leave-One-Out CV**: Each data point is its own fold
- Helps in more reliable model evaluation when data is limited.

Model Evaluation Metrics: Regression Metrics

- **Mean Squared Error (MSE):**

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where y_i is the observed value, $\hat{y}_i$ is the predicted value, and n is the total number of observations.

- **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Gives error in the same unit as the response variable.

- **Coefficient of Determination R^2 :**

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of the observed values. It represents the proportion of variance in y explained by the model.

Model Evaluation Metrics: Classification Metrics

- **Accuracy:**

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

- **Precision and Recall:**

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Precision measures correctness of positive predictions; recall measures ability to find all actual positives.

- **F1 Score** (harmonic mean of precision and recall):

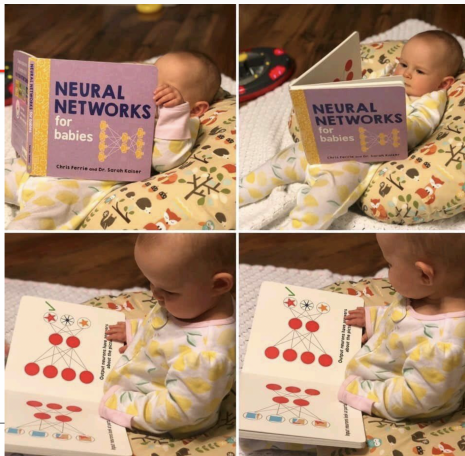
$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Useful when classes are imbalanced or when both precision and recall are important.

Neural Networks

Everybody is learning

Never too late



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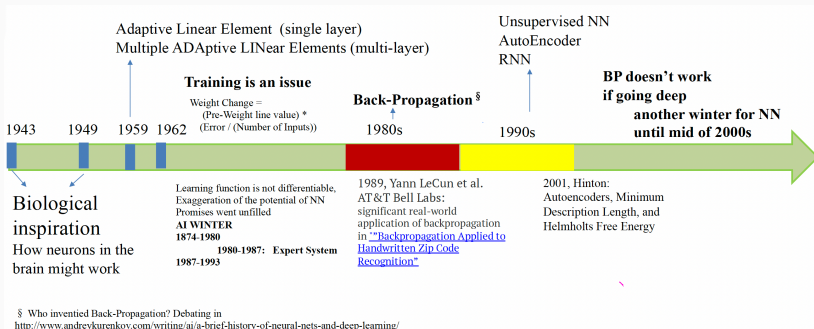
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2.10 Motivation for Deep Neural Network (Deep Learning)

What is a Neural Network?

- A **Neural Network** is a computational model inspired by the way biological neural networks in the human brain process information.
- It consists of **layers** of interconnected **nodes**, or **neurons**, which work together to solve complex problems.
- Similarly, brain activity occurs when a **stimulus** enters the system; information is processed through the network via neurons that extract relevant information, and this information is passed to another area.
- Neural networks can **learn** from data through **training**, making them powerful tools for **pattern recognition** and **predictive modeling**.

History of NNs



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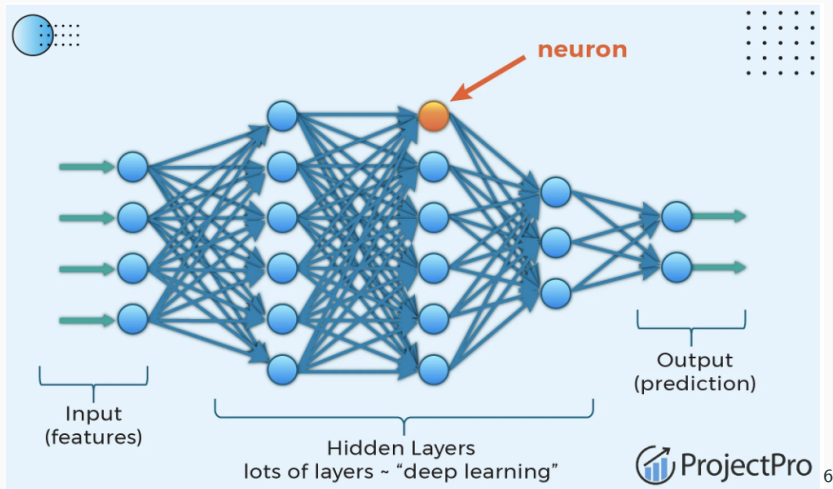
Components of Neural Networks

- Layers
- Neurons
- Weights
- Bias
- Activation Functions
- Forward Propagation
- Loss Function
- Backpropagation
- Learning Rate
- Training Data
- Validation and Test Data

A Basic Architectures of Neural Networks (1)

- A neural network consists of several key components:
- **Input Layer:** Each input represents a single predictor variable, which can be a scalar value, a sequence, an image, or even a sequence of images.
- **Output Layer:** For prediction or classification tasks, this layer typically has a single node. For Conditional Density Estimation (CDE), it contains multiple nodes.
- **Hidden Layers:** These layers, often more than one, perform calculations at each node. Each hidden layer's computations are parameterized by weights and biases.
- The calculations within a layer can vary in type, including standard, convolutional, and recurrent layers.
- The structure of a neural network, referred to as its **architecture**, defines the arrangement and connectivity of these layers.

A Basic Architectures of Neural Networks (2)



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Layers

Neural computing requires a number of neurons, to be connected together into a neural network. Neurons are arranged in layers.

- **Input Layer**

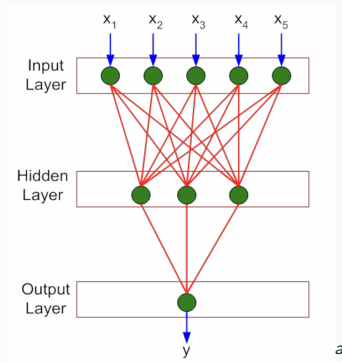
- First layer of neurons.
- Receives raw data directly from the input features.

- **Hidden Layers**

- Layers between the input and output layers.
- Perform computations.
- Can have one or many hidden layers.

- **Output Layer**

- Final layer.
- Produces the network's output.
- Structure depends on the specific task (e.g., classification, regression).



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Neurons, Weights, and Biases

- **Neurons:**

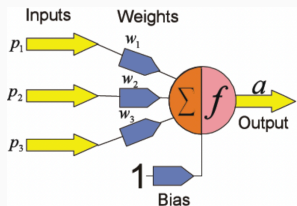
- Basic units of a neural network.
- Receive input, process it, and pass it to other neurons.
- Apply a mathematical function to generate an output.

- **Weights:**

- Parameters that transform input data within the network.
- Each connection between neurons has an associated weight.

- **Biases:**

- Additional parameters in a neuron that allow to fit the data better.
- Provide ability to each neuron have output not 0 when inputs are 0.



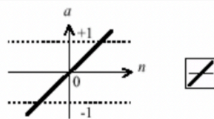
$$a = f(p_1w_1 + p_2w_2 + p_3w_3 + b) = f(\sum p_iw_i + b)$$

Activation function

- The activation function defines the **output of the node**. There is two different functions that we can use, but it must satisfy below two properties
 - **Nonlinear**: If all node has linear activation functions, this is just a linear regression model.
 - **Continuously differentiable**: Needed for gradient based optimization

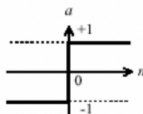
For example, we can think of this activation functions as link function in generalized linear model. In fact a single layered NN is simply a generalized linear model.

Some Examples of Activation Function



$$a = \text{purelin}(n)$$

Linear Transfer Function



$$a = \text{hardlims}(n)$$

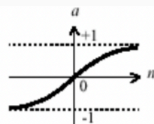
Symmetric Hard Limit Trans. Funct.



ReLU function

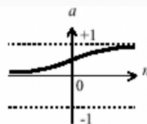


8



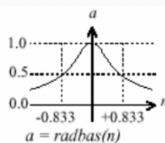
$$a = \text{tansig}(n)$$

Tan-Sigmoid Transfer Function



$$a = \text{logsig}(n)$$

Log-Sigmoid Transfer Function



$$a = \text{radbas}(n)$$

Radial Basis Function



Rectified Linear Unit (ReLU) Activation Function

- Older neural networks relied on **sigmoid** or **tanh** activation functions that suffered from vanishing gradient problems which restrict going to deep
- **ReLU**⁹ activation function somewhat solve this issue and turn out to be a huge help
- **ReLU** has the following forms

$$a(x) = \max\{0, x\}.$$

That means

- The neuron **activates** only when enough information passes through
- **One drawbacks**: Not differentiable at zero.

⁹<https://machinelearningmastery.com/rectified-linear-activation-function-for-deep-learning-neural-networks/>

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2.5 Training of NNs (Learning the Weights)

2.6 Some Important Strategies/Considerations When Training a NN

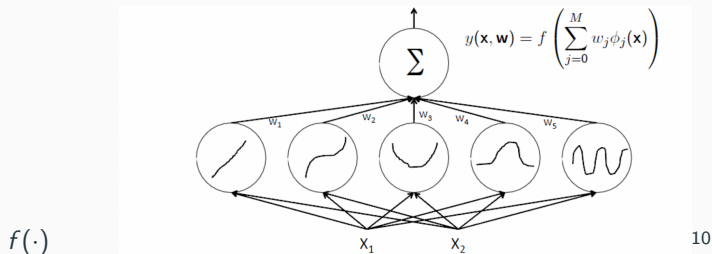
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Linear Regression as Neural Networks



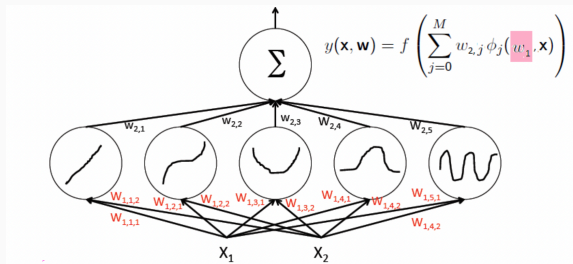
- The identity function in the case of regression.
- A nonlinear activation function in the case of classification.

Learning object:

- Learn $\mathbf{w}$'s so the function fits the data well.
- The basis functions are fixed functions of X .

¹⁰CS 229: Prof. Xiangliang Zhang (KAUST)

Non Linear Regression as Neural Networks



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- Learning object: Learn $\mathbf{w}$ so the function fits the data well
- The basis functions are fixed functions of $\mathbf{x}$ and $\mathbf{w}_1$

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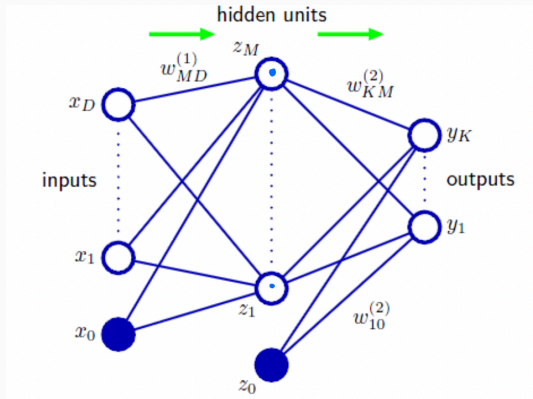
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A Two-Layer Neural Network (1)



- D input variables
- M hidden neurons
- K outputs
- Use the two activation function on weights sum on hidden layers and output layers

A Two-Layer Neural Network (2)

Construct M linear combinations of the inputs $x_1, \dots, x_d$

$$a_j = \sum_{i=0}^D w_{ji}^{(1)} x_i, \quad x_0 = 1.$$

- a_j are the activations, $j = 1, \dots, M$.
- $w_{ji}^{(1)}$ are the layer one weights, $i = 1, \dots, D$.
- $w_{j0}^{(1)}$ are the layer one biases.

Each linear combination a_j is transformed by a (non-linear differentiable) activation function

$$z_j = h(a_j).$$

A Two-Layer Neural Network (3)

The hidden output $z_j = h(a_j)$ are linearly combined in layer two:

$$a_k = \sum_{j=0}^M w_{kj}^{(2)} z_j, \quad z_0 = 1.$$

- a_k are the output activations, $k = 1, \dots, K$.
- $w_{kj}^{(2)}$ are the layer two weights, $j = 1, \dots, M$.
- $w_{k0}^{(2)}$ are the layer one biases.

The output activations a_k are transformed by the output (non-linear differentiable) activation function

$$y_k = \sigma(a_k).$$

- y_k are the final outputs.
- $\sigma(\cdot)$ is like $h(\cdot)$, but often sigmoid function for classification and linear function for Regressions.

A Two-Layer Neural Network (4)

After substituting $y_k = \sigma(a_k)$ the definitions of a_j and a_k :

$$y_k(\mathbf{x}, \mathbf{w}) = \sigma \left(\sum_{i=0}^M w_{kj}^{(2)} h \left(\sum_{i=0}^D w_{ji}^{(1)} x_i \right) \right).$$

Evaluation of this is called **forward propagation**.

- $h(\cdot)$ and $\sigma(\cdot)$ are a sigmoid functions, e.g., the logistics function.

$$s(a) = \frac{1}{1 + \exp(-a)}, \quad s(a) \in [0, 1].$$

- If $\sigma(a)$ is the identity, then a regression model is obtained.

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Recall: Error (Loss) Function

Training a neural network involves estimating the weights and biases by optimizing a loss function. Let N represent the number of observations of the response $t_1, t_2, \dots, t_N$, and let the neural network predict some output $\hat{y}_1, \hat{y}_2, \dots, \hat{y}_N$ (which could be a vector).

- **Regression: Sum of squared**

$$\mathcal{L}(\mathbf{w}) \text{ or } E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(\mathbf{x}_n, \mathbf{w}) - t_n\}^2,$$

where $t_n, n = 1, \dots, N$ is the observed response. N is the total number of training samples (examples).

- **Binary classification:** NNs with only one output node whose activation function is a logistic sigmoid **Cross-entropy error function**

$$\mathcal{L}(\mathbf{w}) \text{ or } E(\mathbf{w}) = -\frac{1}{2} \sum_{n=1}^N \{t_n \log y(\mathbf{x}_n, \mathbf{w}) + (1 - t_n) \log(1 - y(\mathbf{x}_n, \mathbf{w}))\}.$$

Training a Neural Network: Forward and Backward Pass

Each training step has two key phases:

- **Forward Pass**: Compute output predictions from input through the layers.
- **Backward Pass (Backpropagation)**: Compute gradients of loss w.r.t. weights using the chain rule.
- Update weights using **gradient descent** (or a variant).

Goal: Minimize the error/loss function $E(\mathbf{w})$

Forward Propagation: How the Network Makes Predictions

For each layer:

- Compute linear combination (pre-activation):

$$a_j = \sum_i w_{ji} z_i + b_j$$

- Apply activation function:

$$z_j = h(a_j)$$

- Repeat this through hidden layers until final output:

$$\hat{y} = f(\mathbf{x}; \mathbf{w}) = z_{\text{output}}$$

Note:

- z_i : activation from the previous layer (input if it's the first)
- w_{ji} : weight connecting unit i to unit j
- h : activation function (e.g., sigmoid, ReLU, tanh)

Error/Loss Function

At the end of the forward propagation, calculate the loss E , (same as $E(\mathbf{w})$ or $\mathcal{L}(\mathbf{w})$).

- **For Regression (e.g., MSE):**

$$E = \frac{1}{2} \sum_{n=1}^N \|\hat{y}^{(n)} - t^{(n)}\|^2$$

- **For Binary Classification (Cross-Entropy):**

$$E = - \sum_{n=1}^N \left[t^{(n)} \log(\hat{y}^{(n)}) + (1 - t^{(n)}) \log(1 - \hat{y}^{(n)}) \right]$$

- $t^{(n)}$: true target for sample n
 $\hat{y}^{(n)}$: predicted output from forward pass

Backpropagation: Computing the Gradients

For output units:

$$\delta_k = \hat{y}_k - t_k$$

For hidden units:

$$\delta_j = h'(a_j) \sum_k w_{kj} \delta_k$$

Summary of Backward Pass:

- Forward propagate input to compute hidden and output activations.
- Compute output error signal δ_k .
- Backpropagate errors to hidden layers to get δ_j .
- Compute gradient of error w.r.t. weights:

$$\frac{\partial E}{\partial w_{ij}} = \delta_j z_i$$

- Accumulate gradients across samples.

Gradient Descent and Weight Updates

Weight Update Rule:

- Update weights using:

$$w_{ij} \leftarrow w_{ij} - \eta \cdot \frac{\partial E}{\partial w_{ij}}$$

- η : **learning rate**, a small positive number that controls the step size.
- Repeat for multiple **epochs** (complete passes over the dataset) until convergence.

Variants of Gradient Descent:

- **Stochastic Gradient Descent (SGD)**: Updates weights using **one example (data) at a time**. Fast but noisy, often helps escape local minima.
- **Mini-batch Gradient Descent**: Uses a **small batch of samples** (e.g., 32 or 64). Balances efficiency and stability.
- **Adaptive Optimizers (Adam, RMSprop)**: Adjust learning rates **dynamically per parameter**. Often converges faster with minimal tuning.

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Useful Strategies for Training

- **Early Stopping:**
 - Stop training when validation loss stops improving.
- **Dropout Layers:**
 - Randomly deactivate neurons to prevent overfitting.
- **Learning Rate Scheduling:**
 - Decrease learning rate as training progresses.
- **Batch Normalization:**
 - Normalizes activations and accelerates training.

Normalizing Input Data: Why and How?

- **What is Normalization?**

- Rescaling input features to a standard range, e.g., $[0, 1]$ or $\mathcal{N}(0, 1)$.

- **Why is it Important for Neural Networks?**

- Ensures faster and more stable training.
- Prevents activation saturation (e.g., in sigmoid or tanh).
- Keeps gradients in a similar scale across layers → reduces risk of exploding/vanishing gradients.

- **How is it Done?**

- **Standardization:** $x' = \frac{x - \mu}{\sigma}$
- **Min-Max Scaling:** $x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$

What Are Hyperparameters in NNs?

- **Hyperparameters** are settings used to control the learning process in neural networks.
- Unlike model parameters (e.g., weights and biases), they are not learned from data.
- They are set *before* training begins and can significantly affect performance.
- Common examples include:
 - Learning rate (η)
 - Number of layers & units per layer
 - Batch size
 - Number of training epochs
 - Dropout rate, Weight decay, etc.

Key Hyperparameters and Their Role

- **Learning Rate (η):**
 - Controls step size during weight updates.
 - Too high $\rightarrow$ **Divergence**, too low $\rightarrow$ **Slow convergence**.
- **Batch Size:**
 - Number of samples used in one update.
 - Smaller $\rightarrow$ **noisy but faster**; Larger $\rightarrow$ **stable but slower**.
- **Epochs:**
 - One pass over the entire dataset.
 - **Too many** $\rightarrow$ Overfitting; **Too few** $\rightarrow$ Underfitting.

Other Useful Hyperparameters

- **Network Architecture:**
 - Number and size of layers.
 - Affects capacity and generalization.
- **Dropout Rate:**
 - Randomly disables neurons during training.
 - Helps prevent **overfitting**.
- **Weight Decay / L_2 Regularization:**
 - Penalizes large weights.
 - Encourages **simpler models**.

Choosing the Best Model

- Train multiple models with different:
 - Architectures (e.g., number of layers, units)
 - Hyperparameters (learning rate, dropout, etc.)
- Use the **validation set** to select the best model — the one with:
 - **Low validation loss**
 - **Good generalization** (small gap between train and validation loss)
- **Test set** should only be used once — after selection.

Common Pitfalls to Avoid

- Using test set during training → leads to data leakage.
- Evaluating on only one split → consider using **cross-validation**.
- Over-tuning hyperparameters → may overfit to the validation set.
- Ignoring model complexity → simpler models can generalize better.
- Not tracking experiments → use tools like Weights & Biases, TensorBoard, or spreadsheets.

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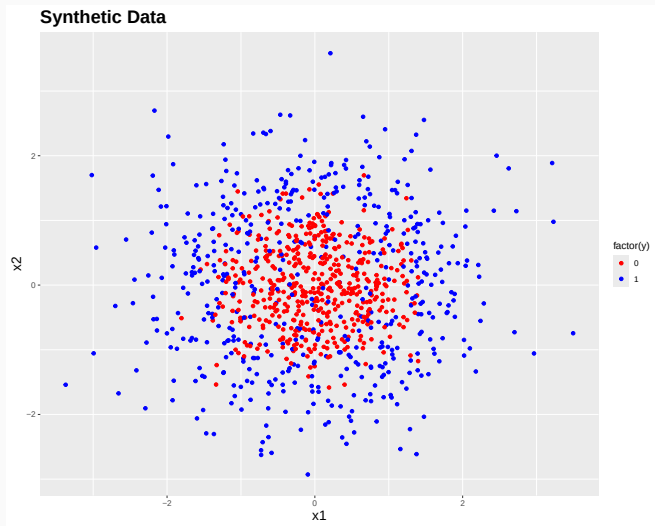
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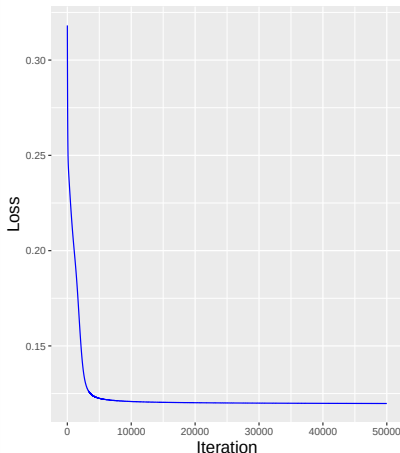
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Lab: Classification Using two-Layer Neural Networks (1)

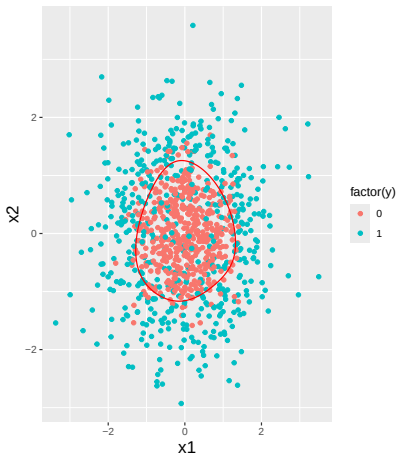


Lab: Classification Using two-Layer Neural Networks (2)

Loss Function Over Iterations



Decision Boundary of the Neural Network



¹²See the python code in the GitHub Link:

https://github.com/yadavrishikesh/Masterclass-IITMandi/blob/main/Module3/code/NNs_for_classification_from_scratch.ipynb

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What is TensorFlow

TensorFlow is an open-source machine learning library developed by Google. It provides a comprehensive, flexible ecosystem of tools, libraries, and community resources that lets researchers push the state-of-the-art in ML, and developers easily build and deploy ML-powered applications.

- Originally developed by the Google Brain Team
- Supports deep learning, numerical computation, and automatic differentiation
- Works across multiple platforms: CPUs, GPUs, TPUs

Typical steps in building a model with TensorFlow:

1. **Prepare** the data
2. **Build** the model (using `keras.Sequential`)
3. **Compile** the model (using `model.compile`) and mention about the loss, optimizer, and metrics
4. **Train** the model (`model.fit`)
5. **Evaluate** the model (`model.evaluate`)
6. **Predict** with new data (`model.predict`)

Why Use TensorFlow?

- **Scalability:** Efficient execution across devices and distributed systems
- **Production Ready:** Tools for deploying models to web, mobile, and edge devices
- **Ecosystem:** Strong support for [visualization](#), [data pipelines](#), and [deployment](#)
- **Community:** Backed by Google and widely adopted in academia and industry

[TensorFlow](#) is designed to support research prototyping and production deployment equally well.

Key Steps in Fitting a Neural Network (TensorFlow)

- **Step 1: Load and Prepare Data**

- Use `tensorflow.keras.datasets` or custom datasets.
- Normalize/scale input features.

- **Step 2: Define the Model Architecture**

- Use `keras.Sequential` or subclassing API.
- Stack layers using `Dense`, `Dropout`, etc.

- **Step 3: Compile the Model**

- Specify `loss function`, `optimizer`, and `metrics`.

Model Definition Example (TensorFlow)

- A simple feedforward network for binary classification:

```
1 model = keras.Sequential([  
2     keras.layers.input(shape=(num_features,)),  
3     keras.layers.Dense(8, activation='relu'),  
4     keras.layers.Dense(10, activation='relu'),  
5     ...  
6     keras.layers.Dense(1, activation='sigmoid')  
7 ])
```

- `input_shape`: Number of input features.
- `activation='relu'`: Adds non-linearity.
- `activation='sigmoid'`: For binary classification.

Compiling the Model

```
1 model.compile(  
2     optimizer='adam',  
3     loss='binary_crossentropy',  
4     metrics=['accuracy']  
5 )
```

- optimizer: Controls weight updates (e.g., adam, sgd).
- loss: Chosen based on task:
 - binary_crossentropy for binary classification
 - categorical_crossentropy for multi-class
 - mse for regression
- metrics: To monitor model performance during training.

Training and Evaluation

```
1 history = model.fit(  
2     X_train, y_train,  
3     validation_data=(X_val, y_val),  
4     epochs=50,  
5     batch_size=32  
6 )
```

- validation_data: Used to monitor generalization.
- epochs: Number of training passes over data.
- batch_size: Mini-batch size.

```
1 test_loss, test_acc = model.evaluate(X_test, y_test)
```

Lab: Fitting the Neural Network using Tensorflow

See the GitHub Link:

https://colab.research.google.com/github/yadavrishikesh/Masterclass-IITMandi/blob/main/Module3/code/NNs_for_classification_using_tensorflow_pac

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A Nice Visual of NNs

Recall: and let's have some fun: <https://playground.tensorflow.org/>

Some key Takeaway: Learning Issue (1)

- Vanishing gradient problems due to propagated error coming from the other neurons and layers
 - Use of smart activation functions solve this issue
- Overfitting due to a lot number of hidden units the number of hidden units determines the complexity of the learned function (the number of parameters $\mathbf{w}$)
 - We may allow regularizations
 - Drop of techniques
 - Sparse neural networks

Some key Takeaway (2): Practical Rule of Thumb on deciding hidden units

- There is no theoretical results that determines the minimum number of hidden units.
- Practical rule of thumb
 - For binary data $M = 2D$
 - For real valued data $M \gg 2D$
- Multiple hidden layers with fewer nodes maybe trained faster for similar quality in some applications

Some Important Comments About the Architecture of NNs

- Typically, K (Output) $< D$ (Input) $< M$ (Hidden).
 - If $M < D$ or $M < K$, information loss may occur at the hidden units.
- A multilayer network of linear units (where all $h(\cdot)$ are linear) is not interesting and can be reduced to a network without hidden units.
- There may be multiple hidden layers.
- Units need not be fully connected to the next layer.
- Links may skip layers.
- There may be symmetries in the weight space, meaning that different choices of w may define the same mapping from input to output.

Tips When Training Neural Networks

- **Always monitor** training and validation loss.
- Use **early stopping** to avoid overfitting.
- **Normalize or standardize** your input features.
- Use **learning rate schedules** or optimizers like Adam.
- Try small models first and increase complexity if needed.
- Perform **hyperparameter tuning** using grid search, random search, or Bayesian optimization.

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Why Go Deeper Than Shallow Networks?

Shallow Neural Networks can approximate any function — *but...*

- Require exponentially many hidden units to model complex functions
- Struggle with high-dimensional input (e.g., images, sequences)
- Lack the ability to learn hierarchical features
- Need heavy feature engineering in many tasks

Deep Neural Networks (DNNs) solve these by composing multiple non-linear transformations.

Success Stories of Deep Learning

- **Computer Vision:** Object detection, facial recognition, medical imaging
- **Natural Language Processing:** Translation, chatbots, summarization
- **Speech & Audio:** Voice assistants, speech-to-text
- **Games & Robotics:** AlphaGo, autonomous driving, robotic arms
- **Healthcare:** Protein folding (AlphaFold), drug discovery

All powered by deep neural networks trained on large datasets and compute.

What's Next: Deep, Adaptive, and Online Learning

Beyond shallow networks:

- **Deep Neural Networks (DNNs)**: Multiple hidden layers; better abstraction
- **Convolutional Neural Networks (CNNs)**: For spatial data like images
- **Recurrent Neural Networks (RNNs)**, **Long Short-Term Memory (LSTMs)** and **Transformers**: For sequences, time series data
- **Online Learning**: Adapt models as data arrives (e.g., stock prediction)
- **Transfer Learning**: Reuse learned models on new tasks (e.g., BERT, ResNet)
- And Many More....

Deep learning opens doors to AI that adapts, understands, and generalizes.